Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
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INSTITUTE OF COMPUTER STUDIES

A WEB-BASED SERVICE MANAGEMENT SYSTEM WITH DATA ANALYTICS USING MACHINE LEARNING ALGORITHMS FOR SHOELOTSKEY VILLAMOR-PASAY

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CHAPTER 1

INTRODUCTION

Small-scale shoe cleaning, restoration, and other service-oriented establishments play an important role in local economic activity, particularly in urban areas where they provide accessible services to the community. However, many small businesses continue to rely on manual or semi-digital processes for managing day-to-day operations. Such practices may make the organization, retrieval, and analysis of operational records more difficult, particularly as transaction records accumulate over time. In an increasingly data-driven business environment, web-based information systems and data analytics provide opportunities for small enterprises to improve record management, operational monitoring, and decision-making.

In the case of Shoelotskey Villamor-Pasay, owned by Ms. Aileen Rosales, the researchers' pre-assessment of the existing operations, supported by consultation with the shop owner and staff and examination of available business records, identified several limitations associated with the existing manual process. Service information and transaction records were primarily maintained through paper-based records, making the retrieval, monitoring, and consolidation of information more time-consuming. The existing process also presented difficulties in monitoring service status, preparing business reports, reviewing historical transactions, and consistently determining expected service completion dates. These operational conditions established the need for a centralized digital system that could organize service records, provide timely access to operational information, and support more systematic business monitoring.

Despite the availability of web-based management systems and machine learning technologies, there remains an opportunity to develop an integrated solution tailored to the operational requirements of a small-scale service establishment such as Shoelotskey. Existing commercial systems may include features or implementation requirements that do not directly correspond to the business's specific workflow and available resources. Furthermore, integrating centralized service management, inventory monitoring, real-time business analytics, and predictive modeling within a single system may provide a more practical approach to addressing the identified operational requirements of the establishment.

To address these challenges, this study proposes the development of a web-based service management system with data analytics using machine learning algorithms for Shoelotskey Villamor-Pasay. The system centralizes core operational processes, including job order management, sales monitoring, inventory tracking, and report generation. It also provides real-time analytics for monitoring business information and incorporates a machine learning model developed using simulated operational data representing verified service configurations and operational conditions, while historical service records were separately used to evaluate the limitations of the original Order-to-Claim prediction target.

Problem Statement

Shoelotskey Villamor-Pasay currently experiences operational challenges associated with its reliance on manual processes. Based on the pre-assessment interview with the business owner and review of available business records, the following problems were identified:

Conceptual Framework

~~conceptual~~ framework

The conceptual framework of the study follows an Input–Process–Output (IPO) model with Evaluation and Feedback components. It illustrates how the required information is processed by the developed system to produce operational outputs, while system quality, user acceptance, machine learning performance, and operational improvements are evaluated separately.

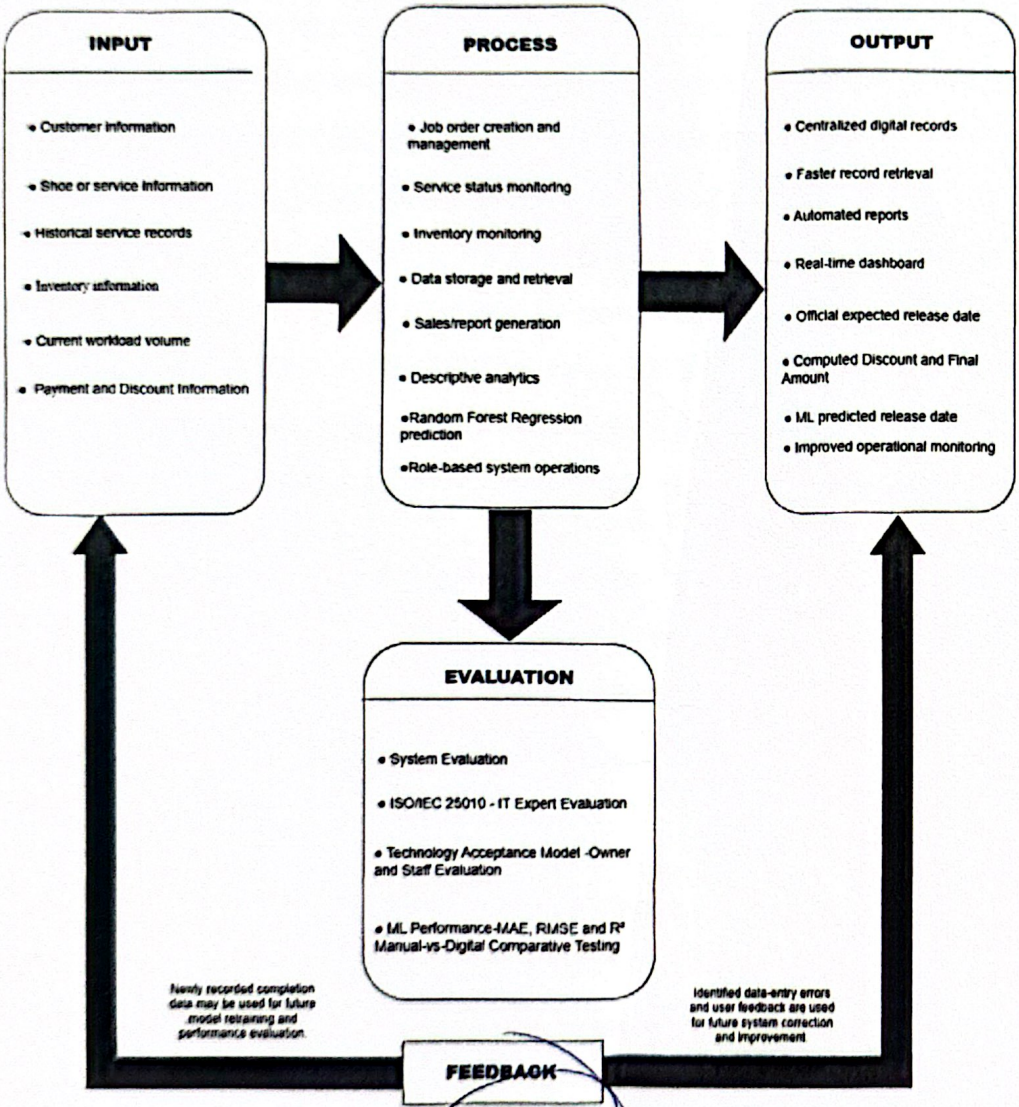


Figure 1: Conceptual Framework

component is evaluated using MAE, RMSE, and R^2, while manual-versus-digital comparative testing is conducted to assess improvements in selected operational processes.

Finally, the Feedback component represents the continuous improvement of the system. Identified data-entry errors and user feedback may be used for future system corrections and enhancements, while newly recorded completion data may be used for future machine learning model retraining and performance evaluation. The feedback loop therefore allows information obtained from system use and evaluation to contribute to subsequent improvements.

Objectives of the Study

This capstone project aims to design and develop the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms in order to improve operational efficiency, data accuracy, service monitoring, and managerial decision-making.

Specific Objectives

Specifically, the project aims to achieve the following:

1. Conduct a comparative analysis between the existing manual paper-based system and the developed centralized digital job order entry and management system in terms of service record accessibility, retrieval time, and accuracy.'
2. Evaluate the predictive performance of the Random Forest Regressor in predicting service turnaround duration and corresponding release dates using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), coefficient of determination (R^2), and prediction tolerance within ±1, ±2, ±3, and ±5 days.

Evaluate the feasibility and limitations of the historical Order-Claim target using actual records, ...

3. Determine the impact of the real-time analytics dashboard on reporting efficiency, accessibility of business information, and decision-making by evaluating report-generation performance and the perceptions of the shop owner and staff regarding the usefulness of the analytics dashboard.
4. Assess the software quality of the developed system based on the IT experts' evaluation using the ISO/IEC 25010 software quality model in terms of:
    - 4.1 Functional suitability;
    - 4.2 Performance efficiency;
    - 4.3 Compatibility;
    - 4.4 Interaction Capability;
    - 4.5 Reliability;
    - 4.6 Security;
    - 4.7 Maintainability;
    - 4.8 Flexibility; and
    - 4.9 Safety).
5. Measure the level of user acceptance of the developed system among the staff and management of Shoelotskey using the Technology Acceptance Model (TAM), measuring perceived usefulness, perceived ease of use, behavioral intention to use, and actual system use.
6. Recommend possible system enhancements by identifying improvements and additional features that may be implemented by future researchers to further enhance system functionality, data analytics capabilities, and user experience.

development. This allowed the researchers to design and implement the service completion date prediction model during the development phase and evaluate it during testing. Overall, the Waterfall methodology provided a clear development structure, supported proper documentation, and ensured that the proposed system met the objectives of the study within the given timeframe.

System Development Framework

The system development framework follows a Three-Tier Architecture, integrating the presentation layer, application layer, and data management layer to support efficient system operation and organized data processing. The development workflow includes requirements gathering, system design, module development, integration testing, and system validation to ensure that the proposed platform meets the operational needs of Shoelotskey.

The system was developed using modern web technologies, including server-side scripting, relational database management systems, and client-side interface frameworks to support web-based access and centralized service management. The layered architecture allows the system to separate user interface functions, system processing logic, and database management, improving maintainability, scalability, and system performance.

Machine learning components are integrated within the application layer to support predictive analytics and service completion estimation. These components analyze historical service records and job order data to generate predicted completion times and identify service trends, assisting administrators in improving operational planning and decision-making.

Analysis of the Existing System

No Likert interpretation such as Acceptable or Highly Acceptable was applied to objective test measurements. Instead, technical test results were evaluated directly against their predefined acceptance criteria.

Population and Sampling Technique

The respondents of this study consisted of individuals directly involved in the operation and evaluation of the proposed system for Shoelotskey. The target population for the internal evaluation consisted of six (6) Shoelotskey personnel, specifically the business owner, co-owner, and four (4) staff who are directly involved in customer transactions, service processing, shoe cleaning and restoration services, and inventory-related activities. These respondents are considered essential because they are the intended end-users of the developed web-based service management system and can provide firsthand feedback regarding its usability, functionality, and overall acceptance.

In addition to the internal respondents, the study included selected Information Technology (IT) experts or Information Technology (IT) practitioners to evaluate the technical quality of the system based on ISO/IEC 25010 standards. These experts may include IT instructors, software developers, system analysts, programmers, or professionals with experience in software evaluation and system development. Their participation is necessary to assess the system in terms of functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety.

The study used total population sampling for the six (6) Shoelotskey personnel because the population is small and manageable. This allows all employees and management personnel to participate in evaluating the system.

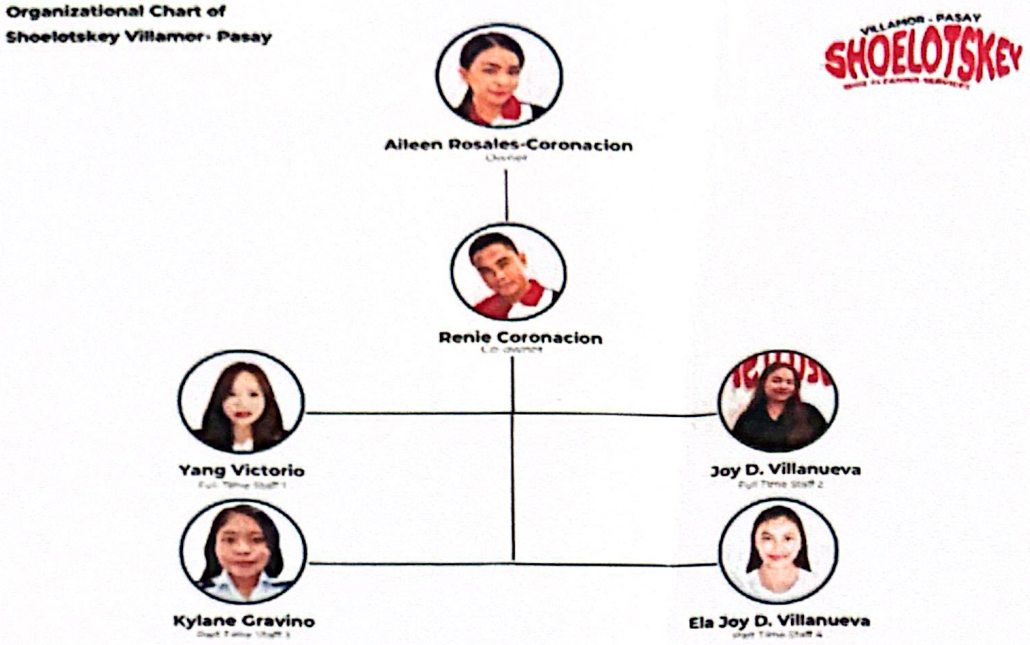


Figure 12. Organizational Chart of Shoelotskey Villamor–Pasay

Figure 12 presents the organizational structure of Shoelotskey Villamor–Pasay. The structure illustrates the relatively small workforce of the establishment and identifies the personnel considered as the intended users of the developed system. All two (2) owners and four (4) staff members were included in the evaluation because they represent the personnel directly involved in the shop's operations and are the intended users of the system. Therefore, the six (6) TAM respondents represent the actual population of qualified users within the establishment rather than a sample drawn from a larger employee population.

who are those ?

In addition to the intended users, five (5) IT experts were purposively selected to evaluate the developed system based on the ISO/IEC 25010 software quality characteristics. The IT experts were selected separately from the shop personnel based on the qualifications established by the researchers, including relevant professional experience and knowledge in information technology and

software systems. Therefore, the study involved a total of eleven (11) respondents: five (5) IT experts and six (6) intended system users.

4.1 Demographic Profile of the Respondents:

Table 16

FREQUENCY AND PERCENTAGE DISTRIBUTION OF RESPONDENTS ACCORDING TO CATEGORY

Category	Frequency	Percentage
IT Expert	5	45.45%
Shop Owner	2	18.18%
Shop Staff	4	36.36%
TOTAL	11	100%

Table 16 shows the frequency and percentage distribution of respondents according to category.

There are 5 or 45.45% respondents are IT Experts. There are 4 or 36.36% are Shop Staff while there are only 2 or 18.18% are Shop Owner.

Findings revealed that the most of the respondents are IT Expert with a frequency of 5 or 45.45%.

4.2 Comparative Analysis of Manual and Developed System

The comparative analysis evaluated the existing manual paper-based process against the developed centralized digital system. The comparison focused on service record accessibility, retrieval time, and data accuracy. These indicators were selected because they directly correspond to the operational challenges identified in Chapter 1 and the first specific objective of the study.